

DATA VISUALIZATION & CAPSTONE PROJECT

Final Project Explanation Report

Project Focus

Financial Performance, Sales & Profitability Analysis

Tools

Python / Pandas • Excel • Power BI • SQL/Pandas Analysis

Report Purpose

To explain the analysis process, dashboard insights, key findings and business recommendations in a concise stakeholder-friendly format.

Prepared for Week 4 Capstone Project submission

1. Executive Summary

This project analyzes financial and sales performance across products, countries, customer segments and time. The objective was to turn a raw business dataset into a clear analytical story that a stakeholder can use to understand where revenue is being generated, where profit is concentrated and which areas deserve closer attention.

The analysis shows a strong overall revenue base of **\$118.73M** and total profit of **\$16.89M**. That corresponds to an overall profit margin of approximately **14.23%** (profit divided by sales). The dashboard also records approximately **1.13M units sold** and an average sales price of **\$118.43**.

A key distinction in the analysis is that the strongest market by sales is not necessarily the strongest market by profit. The United States of America leads on sales, while France is the leading country on profit. Paseo is the leading product by profit, and the Government segment is the dominant contributor to overall profitability.

\$118.73M	\$16.89M	1.13M	\$118.43
Total Sales	Total Profit	Units Sold	Average Sales Price

2. Project Methodology

The project followed an end-to-end analytics workflow:

- **Data preparation:** The dataset was inspected and prepared for analysis, with attention to data types, missing values, duplicates and usable analytical fields.
- **Exploratory analysis:** Sales, profit, units, products, countries and segments were examined to identify performance patterns.
- **Business analysis:** Aggregations and comparisons were used to identify high-performing and high-profit areas.
- **Visualization:** The results were presented through Excel/Power BI visuals and KPI cards so that stakeholders can move from an overall view to detailed performance.
- **Interpretation:** The final step was to translate the numbers into concise findings and practical business recommendations.

3. Dashboard & KPI Interpretation

The dashboard is designed around four headline measures: sales, profit, units sold and average sales price. Together, these metrics provide a balanced view of scale and financial outcome rather than relying on sales alone.

Sales vs. Profit: \$118.73M in sales produced \$16.89M in profit, giving an overall margin of about 14.23%. This indicates that the dashboard should be read through both revenue and profitability lenses.

Volume: Approximately 1.13M units were sold. This provides context for the revenue figure and helps distinguish high-volume performance from performance driven mainly by price.

Average sales price: The average sales price is \$118.43 per unit. This metric can be monitored alongside units and profit to understand whether changes in revenue are being driven by volume, pricing or product mix.

4. Key Performance Findings

4.1 Geographic Performance

United States of America – highest sales: The United States is the leading country by sales in the dashboard. This makes it an important revenue market and a natural focus for understanding demand, customer mix and product performance.

France – highest profit: France is the strongest country by profit. The difference between the sales leader and profit leader is important: a market can generate substantial revenue without generating the highest profit. This suggests that pricing, discounting, product mix or cost structure may be influencing country-level profitability.

4.2 Segment Performance

Government – leading profit contributor: The Government segment contributes the largest share of profit. This makes the segment strategically important from a profitability perspective. Its performance should therefore be monitored not only for sales volume but also for margin, product mix and discount levels.

4.3 Product Performance

Paseo – leading product by profit: Paseo is the top-performing product on profitability in the analysis. Its position suggests that the product has a meaningful contribution to the overall bottom line. Performance of other leading products should be compared against Paseo to understand whether its advantage comes from sales volume, pricing, lower discounts or a combination of factors.

4.4 Time-Based Performance

The dashboard includes time-based views so that stakeholders can identify monthly or yearly changes in sales and profit. The time trend should be interpreted together with product, country and segment filters because an overall monthly movement may be driven by a particular market or customer group.

5. Stakeholder Insights & Recommendations

- 1. Overall performance:** The business generated \$118.73M in sales and \$16.89M in profit, with approximately 1.13M units sold and an average sales price of \$118.43.
- 2. Sales market:** The United States of America recorded the highest sales, making it an important market for maintaining and growing revenue.
- 3. Profit market:** France recorded the highest profit, showing that the market generating the most revenue is not necessarily the market generating the most profit.
- 4. Segment:** Government is the leading profit-contributing segment and is therefore a key area for understanding and protecting profitability.
- 5. Product:** Paseo is the leading product by profit and should be monitored as an important contributor to the overall financial result.

Recommendations

- Continue strengthening the United States market while monitoring its profitability, not only its sales volume.
- Study the factors behind France's high profit performance—particularly product mix, pricing and discount levels—and assess whether similar practices can be applied elsewhere.
- Maintain strong relationships with Government customers and monitor their product mix and profit margins.
- Protect the profitability of Paseo while comparing its pricing, discount and sales-volume drivers with other major products.
- Use monthly and yearly trend views regularly to identify periods of growth, slowdown or unusual profitability and then drill down by country, segment and product.

6. Conclusion

The project converts business data into a stakeholder-oriented view of financial performance. The overall results show a substantial sales base and positive profitability, while the detailed analysis highlights an important difference between where sales are generated and where profit is generated. The United States is the leading sales market, France leads on profit, Government is the strongest profit-contributing segment, and Paseo is the leading product by profit.

The dashboard therefore provides more than a snapshot of revenue: it helps stakeholders identify the markets, customer groups and products that contribute most to financial performance and provides a basis for deeper investigation of pricing, discounting and product-mix decisions.

7. Week 4 Deliverables

Deliverable	Submission / Purpose
Complete Code	GitHub repository link or ZIP containing the project code/notebook and supporting analysis files.
PDF Explanation Report	This report: project methodology, KPI interpretation, dashboard insights, findings and Recommendation.
Dashboard	Power BI/Tableau dashboard used to visualize business metrics and support stakeholder interpreter

Note: The report uses the confirmed dashboard metrics and findings from the project analysis. Exact visual screenshots can be appended to the dashboard section when the final Power BI dashboard image is available.